Bank & SSC Quant formula sheet

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Percentages

- Net change of a% then b% = a + b + ab/100 %
  Use negative values for falls.
- Rise to recover a fall of x% = x/(100 − x) × 100 %
- Price ↑ r% ⇒ cut use by r/(100 + r) × 100 %
  Same spending.

Profit, loss & discount

- Profit % = (SP − CP)/CP × 100 Always on cost price.
- SP = MP × (1 − d/100) Discount is on marked price.
- Successive discounts d_1, d_2 = $d_1 + d_2 - d_1d_2/100$ %
- Same SP, x% gain and x% loss ⇒ loss of $x^2/100$ %

Interest

- SI = PRT/100
- $A = P(1 + R/100)^n$
- CI − SI (2 years) = $P(R/100)^2$
- CI − SI (3 years) = $P(R/100)^2(3 + R/100)$
- Half-yearly: rate R/2, periods 2n

Speed & work

- km/h × 5/18 = m/s
- Average speed (equal distances) = 2xy/(x + y)
- Boat b = (down + up)/2, stream s = (down − up)/2
- A alone a days, B alone b days ⇒ together ab/(a + b) days
- $M_1D_1H_1/W_1 = M_2D_2H_2/W_2$ Men × days × hours per unit of work.

Averages & mixtures

- Average = sum/count
- Alligation: cheaper : dearer = (d − m) : (m − c)
- After n replacements of fraction f: left = $(1 - f)^n$ of original

Mensuration

- Rectangle area lw, perimeter 2(l + w)
- Trapezium area = ½(a + b)h
- Rhombus area = $½d_1d_2$
- Cube V = a^3, S = $6a^2$; cuboid S = 2(lb + bh + hl)
- Cylinder CSA = 2πrh, TSA = 2πr(r + h)
- Cone slant l = $\sqrt{(r^2 + h^2)}$, CSA = πrl